A Population objective and possible estimation bias

A.1 Population-objective identity

Using the notation of Appendix D, let

$$\mathcal{J}(\theta) = \frac{1}{2}\mathbb{E}[r^2_{\nu,\mathrm{FSM}}(t;\theta)] \tag{A.1}$$

denote the population counterpart of the innovation-energy criterion. The following identity is stated for stationary filtering and an exact representation of the true signal path, under the assumptions of that appendix. Define

$$\Delta\beta(t;\theta) = \beta_{\mathrm{FSM}}(t;\theta^\star) - \beta_{\mathrm{FSM}}(t;\theta), \tag{A.2}$$

$$d = [d_1, \ldots, d_{n_d}]^\top, \qquad E^\star(t) = [e^\star(t-1), \ldots, e^\star(t-n_d)]^\top,$$

$$\Gamma_e = \mathbb{E}[E^\star(t)E^\star(t)^\top]. \tag{A.3}$$

The residual decomposition established there gives

$$2\mathcal{J}(\theta) = \sigma_\nu^2 + (d-d^\star)^\top\Gamma_e(d-d^\star) + \mathbb{E}\Big[\big(D(z^{-1};\theta)\Delta\beta(t;\theta)\big)^2\Big]. \tag{A.4}$$

The cross terms vanish because the input is independent of the innovation process and $\nu(t)$ is orthogonal to past disturbances. Consequently, under identifiability and $\Gamma_e \succ 0$, the true parameter $\theta^\star$ uniquely minimizes the population objective. This statement concerns fixed candidate parameters; it does not imply that estimates obtained from a finite noisy record are unbiased. For a fixed FSM approximation that does not represent the true signal path exactly, an additional approximation error must be considered.

A.2 Reconstructed regressors and statistical bias

In the PL formulation, the unmeasured internal signals and disturbances are replaced by their reconstructions to build $\hat{\Phi}^{(k)}$. These reconstructions and the innovation estimates are updated using the same noisy output record. Their estimation errors can therefore share noise with the regression residual, so the usual regressor–error orthogonality need not hold. Regressor contamination and the resulting bias of ordinary least squares are standard errors-in-variables issues in system identification [3].

To see the mechanism, let $\varphi^\star(t)^\top$ be a row of the ideal PL regressor and $\hat{\varphi}(t)^\top$ its reconstructed counterpart at a given outer iteration. Since $c(t) = \varphi^\star(t)^\top\theta^\star + \nu(t)$, the regression using the reconstructed signals is

$$c(t) = \hat{\varphi}(t)^\top\theta^\star + \varepsilon(t), \tag{A.5}$$

$$\varepsilon(t) = \nu(t) - [\hat{\varphi}(t) - \varphi^\star(t)]^\top\theta^\star. \tag{A.6}$$

The reconstruction error enters both $\hat{\varphi}(t)$ and $\varepsilon(t)$. Hence, even though $\nu(t)$ is white, the effective regression error need not be orthogonal to the reconstructed regressor. If this correlation is present, least-squares updates can yield statistically biased estimates.

Related block-oriented identification methods address this issue using instruments constructed from noise-free signal counterparts, together with iterative reconstruction of unmeasured internal signals [2]. Janczak and Korbicz [1] give an explicit example in which regressor–disturbance correlation makes a transformed Wiener least-squares estimator inconsistent. Their disturbance placement differs from the output-noise model considered here, so that result illustrates the mechanism rather than proving bias of the present algorithms.

Joint estimation alone does not establish bias. In particular, dependence on past colored disturbances does not imply correlation with the current white innovation. The relevant question is whether the implemented reconstruction preserves orthogonality with its effective regression error. This must be assessed for the estimator actually used; the population identity alone does not settle its finite-record bias.

A.3 Record-length study

[Results and discussion will be inserted after the record-length study is complete.]

References

[1] Andrzej Janczak and Józef Korbicz. Two-stage instrumental variables identification of polynomial Wiener systems with invertible nonlinearities. *International Journal of Applied Mathematics and Computer Science*, 29(3):571–580, 2019.

[2] Boyi Ni, Marion Gilson, and Hugues Garnier. Refined instrumental variable method for Hammerstein–Wiener continuous-time model identification. *IET Control Theory & Applications*, 7(9):1276–1286, 2013.

[3] Torsten Söderström. Errors-in-variables methods in system identification. *Automatica*, 43(6):939–958, 2007.